

# Example: Replication in the binomial model

Computational Methods in Finance, Code 21040

Anna Battauz

Bocconi University

# The Binomial Model

## Securities:

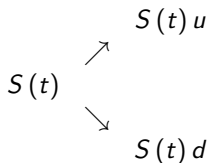
- The *riskless* security with price  $B = \{B(t)\}_{t=0}^T$ , whose price at time  $t$  is known at  $t - \Delta t$ . We also assume  $B(0) = 1$ .

The return

$$r\Delta t = \frac{B(t + \Delta t) - B(t)}{B(t)} \text{ with constant } r$$

for  $t = 0, \Delta t, \dots, T - \Delta t$

- The *risky* security with prices  $S = \{S(t)\}_{t=0}^T$ , with  $d < u$



# The Risk Neutral Probability

$\mathbb{Q}(up) = q$  and  $\mathbb{Q}(down) = 1 - q$  such that

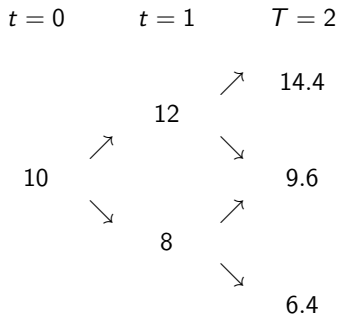
$$\begin{aligned}\mathbb{E}^{\mathbb{Q}} \left[ \frac{S(t + \Delta t) - S(t)}{S(t)} \middle| \mathcal{P}_t \right] &= r\Delta t \\ \frac{S(t)u - S(t)}{S(t)}q + \frac{S(t)d - S(t)}{S(t)}(1 - q) &= r\Delta t \\ (u - 1)q + (d - 1)(1 - q) &= r\Delta t \\ (u - d)q + (d - 1) &= r\Delta t \\ q &= \frac{1 + r\Delta t - d}{(u - d)} \\ 1 - q &= \frac{u - (1 + r\Delta t)}{(u - d)}\end{aligned}$$

$$NA \iff d < 1 + r\Delta t < u$$

## Example

$$\begin{aligned}T &= 2 \quad \Delta t = 1 \quad r = 10\% \\S(0) &= 10 \quad u = 1.2 \quad d = 0.8 \\q &= \frac{1 + r - d}{u - d} = 0.75\end{aligned}$$

Tree for values of  $S$  :



# European Put Option on $S$

Strike  $K = 10$  and maturity  $T = 2$ .

$V_E(t)$  = Value of the European put option at  $t$  is

$$V_E(2) = X(2) = (K - S(2))^+$$

$$V_E(1) = \mathbb{E}^{\mathbb{Q}} \left[ \frac{V_E(2)}{1+r} \middle| \mathcal{P}_1 \right]$$

$$V_E(0) = \mathbb{E}^{\mathbb{Q}} \left[ \frac{V_E(1)}{1+r} \right]$$

$t = 0$  $t = 1$  $T = 2$ 

$$S(0) = 10$$
$$V_E(0) = \frac{0.09 \cdot 0.75 + 1.09 \cdot 0.25}{1.1} = 0.31$$

$$S(1) = 12$$
$$V_E(1) = \frac{0.09 \cdot 0.75 + 0.4 \cdot 0.25}{1.1} = 0.09$$

$$S(1) = 8$$
$$V_E(1) = \frac{0.4 \cdot 0.75 + 3.6 \cdot 0.25}{1.1} = 1.09$$

$$S(2) = 14.4$$
$$V_E(2) = X(2) = 0$$

$$S(2) = 9.6$$
$$V_E(2) = X(2) = 0.4$$

$$S(2) = 6.4$$
$$V_E(2) = X(2) = 3.6$$

## Replicating the European put

At time  $t = 1$ , we select  $\vartheta(1)$  such that  $V_{\vartheta}(2) = X(2)$ . This gives

$$\begin{bmatrix} 1.1^2 & 14.4 \\ 1.1^2 & 9.6 \end{bmatrix} \begin{bmatrix} \vartheta_0(1) \\ \vartheta_1(1) \end{bmatrix} = \begin{bmatrix} 0 \\ 0.4 \end{bmatrix} \quad \text{if } S(1) = 12$$

$$\begin{bmatrix} 1.1^2 & 9.6 \\ 1.1^2 & 6.4 \end{bmatrix} \begin{bmatrix} \vartheta_0(1) \\ \vartheta_1(1) \end{bmatrix} = \begin{bmatrix} 0.4 \\ 3.6 \end{bmatrix} \quad \text{if } S(1) = 8$$

$$\begin{bmatrix} \vartheta_0(1) \\ \vartheta_1(1) \end{bmatrix} = \begin{cases} \begin{bmatrix} 1.1^2 & 14.4 \\ 1.1^2 & 9.6 \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ 0.4 \end{bmatrix} = \begin{bmatrix} 0.99174 \\ -8.3 \times 10^{-2} \end{bmatrix} & \text{if } S(1) = 12 \\ \begin{bmatrix} 1.1^2 & 9.6 \\ 1.1^2 & 6.4 \end{bmatrix}^{-1} \begin{bmatrix} 0.4 \\ 3.6 \end{bmatrix} = \begin{bmatrix} 8.2645 \\ -1.0 \end{bmatrix} & \text{if } S(1) = 8 \end{cases}$$

The cost of  $\vartheta$  at  $t = 1$  is

$$V_{\vartheta}(1) = \begin{cases} 0.99174 \cdot 1.1 - 8.3333 \times 10^{-2} \cdot 12 = 0.09 & \text{if } S(1) = 12 \\ 8.2645 \cdot 1.1 - 1.0 \cdot 8 = 1.09 & \text{if } S(1) = 8 \end{cases}$$

that coincides with  $V_E(1)$ .

## Replicating the European put

At time  $t = 0$ , we select  $\vartheta(0)$  such that the value of  $\vartheta(0)$  at  $t = 1$  equals  $V_{\vartheta}(1) = V_E(1)$ .

This gives

$$\begin{bmatrix} 1.1 & 12 \\ 1.1 & 8 \end{bmatrix} \begin{bmatrix} \vartheta_0(0) \\ \vartheta_1(0) \end{bmatrix} = \begin{bmatrix} 0.09 \\ 1.09 \end{bmatrix}$$

$$\begin{bmatrix} \vartheta_0(0) \\ \vartheta_1(0) \end{bmatrix} = \begin{bmatrix} 1.1 & 12 \\ 1.1 & 8 \end{bmatrix}^{-1} \begin{bmatrix} 0.09 \\ 1.09 \end{bmatrix} = \begin{bmatrix} 2.8091 \\ -0.25 \end{bmatrix}$$

whose value at  $t = 0$  is

$$V_{\vartheta}(0) = 2.809 \cdot 1 + (-0.25) \cdot 10 = 0.31 = V_E(0).$$

# American Put Option on $S$

Strike  $K = 10$  and maturity  $T = 2$ .

$V(t)$  = Value of the American put option at  $t$  is

$$V(2) = X(2) = (K - S(2))^+$$

$$V(1) = \max(X(1), CV(1)) \quad \text{with } CV(1) = \mathbb{E}^{\mathbb{Q}} \left[ \frac{V(2)}{1+r} \middle| \mathcal{P}_1 \right]$$

$$V(0) = \max(X(0), CV(0)) \quad \text{with } CV(0) = \mathbb{E}^{\mathbb{Q}} \left[ \frac{V(1)}{1+r} \right]$$

# American Put Option on S

$t = 0$

$t = 1$

$T = 2$

$$\begin{aligned} S(0) &= 10 & X(0) &= 0 \\ CV(0) &= \frac{0.09 \cdot 0.75 + 2 \cdot 0.25}{1.1} = 0.51 \\ V(0) &= \max(0, 0.51) = 0.51 \end{aligned}$$

$$\begin{aligned} S(1) &= 12 & X(1) &= 0 \\ CV(1) &= \frac{0 \cdot 0.75 + 0.4 \cdot 0.25}{1.1} = 0.09 \\ V(1) &= \max(0, 0.09) = 0.09 \end{aligned}$$

$$\begin{aligned} S(1) &= 8 & X(1) &= 2 \\ CV(1) &= \frac{0.4 \cdot 0.75 + 3.6 \cdot 0.25}{1.1} = 1.09 \\ V(1) &= \max(1.09, 2) = 2 \end{aligned}$$

$$\begin{aligned} S(2) &= 14.4 \\ V(2) &= X(2) = 0 \end{aligned}$$

$$\begin{aligned} S(2) &= 9.6 \\ V(2) &= X(2) = 0.4 \end{aligned}$$

$$\begin{aligned} S(2) &= 6.4 \\ V(2) &= X(2) = 3.6 \end{aligned}$$

## Replicating the American put

At  $t = 1$  select  $\vartheta(1) = \begin{bmatrix} \vartheta_0(1) \\ \vartheta_1(1) \end{bmatrix}$  such that  $V_{\vartheta}(2) = V(2) = X(2)$  :

$$\vartheta_0(1) B(2) + \vartheta_1(1) S(2) = V(2)$$

At  $t = 0$  select  $\vartheta(0) = \begin{bmatrix} \vartheta_0(0) \\ \vartheta_1(0) \end{bmatrix}$  such that the value at  $t = 1$  of  $\vartheta(0)$  is equal to  $V(1)$  :

$$\vartheta_0(0) B(1) + \vartheta_1(0) S(1) = V(1)$$

## Replicating the American put

At time  $t = 1$ , we select  $\vartheta(1)$  such that  $V_{\vartheta}(2) = X(2)$ . This gives

$$\begin{bmatrix} 1.1^2 & 14.4 \\ 1.1^2 & 9.6 \end{bmatrix} \begin{bmatrix} \vartheta_0(1) \\ \vartheta_1(1) \end{bmatrix} = \begin{bmatrix} 0 \\ 0.4 \end{bmatrix} \quad \text{if } S(1) = 12$$

$$\begin{bmatrix} 1.1^2 & 9.6 \\ 1.1^2 & 6.4 \end{bmatrix} \begin{bmatrix} \vartheta_0(1) \\ \vartheta_1(1) \end{bmatrix} = \begin{bmatrix} 0.4 \\ 3.6 \end{bmatrix} \quad \text{if } S(1) = 8$$

$$\begin{bmatrix} \vartheta_0(1) \\ \vartheta_1(1) \end{bmatrix} = \begin{cases} \begin{bmatrix} 1.1^2 & 14.4 \\ 1.1^2 & 9.6 \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ 0.4 \end{bmatrix} = \begin{bmatrix} 0.99174 \\ -8.3 \times 10^{-2} \end{bmatrix} & \text{if } S(1) = 12 \\ \begin{bmatrix} 1.1^2 & 9.6 \\ 1.1^2 & 6.4 \end{bmatrix}^{-1} \begin{bmatrix} 0.4 \\ 3.6 \end{bmatrix} = \begin{bmatrix} 8.2645 \\ -1.0 \end{bmatrix} & \text{if } S(1) = 8 \end{cases}$$

The cost of  $\vartheta$  at  $t = 1$  is

$$V_{\vartheta}(1) = \begin{cases} 0.99174 \cdot 1.1 - 8.3333 \times 10^{-2} \cdot 12 = 0.09 & \text{if } S(1) = 12 \\ 8.2645 \cdot 1.1 - 1.0 \cdot 8 = 1.09 & \text{if } S(1) = 8 \end{cases}$$

and

$$V_{\vartheta}(1) = \begin{cases} 0.09 = V(1) & \text{if } S(1) = 12 \\ 1.09 < 2 = V(1) & \text{if } S(1) = 8 \end{cases}$$

## Replicating the American put

At time  $t = 0$ , we select  $\vartheta(0)$  such that the value of  $\vartheta(0)$  at  $t = 1$  equals  $V(1)$ . This gives

$$\begin{bmatrix} 1.1 & 12 \\ 1.1 & 8 \end{bmatrix} \begin{bmatrix} \vartheta_0(0) \\ \vartheta_1(0) \end{bmatrix} = \begin{bmatrix} 0.09 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} \vartheta_0(0) \\ \vartheta_1(0) \end{bmatrix} = \begin{bmatrix} 1.1 & 12 \\ 1.1 & 8 \end{bmatrix}^{-1} \begin{bmatrix} 0.09 \\ 2 \end{bmatrix} = \begin{bmatrix} 5.2909 \\ -0.4775 \end{bmatrix}$$

whose value at  $t = 0$  is

$$V_{\vartheta}(0) = 5.2909 \cdot 1 + (-0.4775) \cdot 10 = 0.51 = V(0).$$

## Replicating the American put

When is the writer of the option allowed to consume? It depends on the exercise decision of the holder. The optimal exercise policy in this example is

$$\tau^* = \begin{cases} 2 & \text{if } S(1) = 12 \\ 1 & \text{if } S(1) = 8 \end{cases}$$

If this policy is adopted by the holder of the option, then the writer uses all the initial money to hedge his position.

If the option holder *does not optimally exercise* the option at  $t = 1$  when  $S(1) = 8$ , then the writer can consume

$$V(1) - CV(1) = 2 - 1.09 = 0.91,$$

by liquidating the positions  $\vartheta(0)$  at the value  $V(1)$ , and entering in the cheaper new positions  $\vartheta(1)$ , whose cost is  $CV(1)$ .